

Atakan Aygün

PhD Candidate, Mechanical Engineering

Middle East Technical University, Mechanical Engineering G-157, 06800 Ankara, Türkiye

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Education

Middle East Technical University , PhD in Mechanical Engineering	2023 – present
• Supervisors: Dr. Ali Karakus & Dr. Tim Warburton	
Middle East Technical University , MSc in Mechanical Engineering	2020 – 2023
• Supervisors: Dr. Ali Karakus & Dr. Romit Maulik	
• Thesis: Physics-Informed Neural Networks for Computational Fluid Dynamics	
Middle East Technical University , BSc in Mechanical Engineering	2015 – 2020

Employment

Research Assistant , METU, Mechanical Engineering Department	2022 – present
Part-time Researcher , TÜBİTAK SAGE	2020

Academic Visits

The University of Texas at Austin , Oden Institute for Computational Engineering and Sciences, USA	October 2026 – May 2027
• Host: Dr. Jesse Chan	
• Funded by TUBITAK 2214 International Doctoral Research Fellowship	
University of Stuttgart , Institute of Aerodynamics and Gasdynamics, Germany	July 2025
• Host: Dr. Andrea Beck, Numerics Research Group	
• Research visit on GPU-accelerated ALE discontinuous Galerkin methods for moving domain problems	

Publications

Preprints In Review

- Aygün A.**, Ata O., Warburton T., Karakus A. (2026). A High-Order Arbitrary Lagrangian-Eulerian Discontinuous Galerkin Method for the Boltzmann Equation in Nearly Incompressible Flows.
- Kaya F., Koc B., **Aygün A.**, Ata O., Karakus A. (2026). Hybrid ROM-PINN Framework for Closure Modeling in Convection-Dominated Systems.

Peer-Reviewed Journal Articles

- Ata O., **Aygün A.**, Warburton T., Karakus A. (2026). Implicit large eddy simulation of nearly incompressible flows with a discontinuous Galerkin–Boltzmann formulation. *Physics of Fluids*, 38(5), 055134.
- Aygün A.**, Karakus A. (2024). Physics-informed neural networks for weakly compressible flows using Galerkin–Boltzmann formulation. *Physics of Fluids*, 36(11), 117125.
- Aygün A.**, Maulik R., Karakus A. (2023). Physics-informed neural networks for mesh deformation with exact boundary enforcement. *Engineering Applications of Artificial Intelligence*, 125, 106660.
- Aygün A.**, Karakus A. (2022). Physics-informed neural networks for two dimensional incompressible thermal convection problems. *Journal of Thermal Sciences and Technology*, 42(2), 221–232.

Conference Proceedings and Talks

1. **Aygün A.**, Ata O., Karakus A., GPU Accelerated Discontinuous Galerkin Solutions of Galerkin-Boltzmann Formulations for Nearly Incompressible Flows on Moving Meshes, *SIAM Conference on Parallel Processing for Scientific Computing (PP26)*, 2026, Berlin, Germany
2. **Aygün A.**, Ata O., Karakus A., Physics-Informed Neural Networks for Heat Transfer and Fluid Flow Problems, *15th International Conference on Computational Heat and Mass Transfer*, 2025, Antalya, Türkiye
3. **Aygün A.**, Karakus A., Discontinuous Galerkin Solutions of Moving Boundary Problems Using ALE Method on GPUs, *SIAM CSE25*, 2025, Fort Worth, Texas, USA
4. Taşdelen A. S., **Aygün A.**, Karakus A., Remin: A Physics-Informed Neural Network Framework and Its Application to Thermal Convection Problems, *24th Congress on Thermal Science and Technology*, 2023, Ankara, Türkiye
5. **Aygün A.**, Karakus A., Physics-Informed Neural Networks for Mesh Deformation with Exact Boundary Enforcement, *SIAM CSE23*, 2023, Amsterdam, Netherlands
6. **Aygün A.**, Karakus A., Physics-Informed Neural Networks to Solve 1D Euler Equations, *National Conference of Aerospace*, 2022, İzmir, Türkiye (Virtual)

Grants and Scholarships

- 2026 - TUBITAK 2214 International Research Fellowship for funding the research in Oden Institute at The University of Texas at Austin.
- 2026 - HPC Allocation (PI): “GPU-accelerated DG Boltzmann methods for nearly incompressible flows on moving domains”, Barcelona Supercomputing Center, 3000 node hours on MareNostrum 5 accelerated partition.
- 2026 - METU Travel Grant to attend SIAM Conference on Parallel Processing 26 in Berlin, Germany.
- 2025 - Erasmus+ KA131 Short Term Doctoral Traineeship Mobility grant to visit University of Stuttgart.
- 2025 - TUBITAK 2224 Travel Grant to attend SIAM Conference on Computational Science and Engineering 25 in Fort Worth, TX, USA.
- 2023 - TUBITAK 2224 Travel Grant to attend SIAM Conference on Computational Science and Engineering 23 in Amsterdam, The Netherlands.
- 2023-2026 - TUBITAK 2211 National PhD Scholarship.

Review Activity

- Computer Physics Communications (Elsevier)
- Physics of Fluids (AIP)
- Applied Mathematical Modelling (Elsevier)
- The European Physical Journal Plus (Springer)
- International Journal for Numerical Methods in Engineering (Wiley)

Teaching Assistant Experience

Middle East Technical University

- ME210 – Applied Mathematics for Mechanical Engineers (Spring 2022, 2023, 2024, 2026)
- ME305 & ME306 – Fluid Mechanics I & II (Fall 2024; Spring 2023, 2025)
- ME310 – Numerical Methods (Fall 2022, 2024; Spring 2026)
- ME311 – Heat Transfer (Fall 2025)
- ME351 – Thermodynamics of Heat Power (Spring 2025)
- ME484 – Turbulence and Its Measurement (Spring 2023)
- ME485 – Computational Fluid Dynamics Using Finite Volume Method (Fall 2024)
- ME489 – Applied Scientific Programming (Fall 2024, 2026)